Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 8: Working with Fractions

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	8 — Working with Fractions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Convert every mixed number to an improper fraction before multiplying or dividing, flip only the divisor, remember 'of the remaining' always acts on the leftover and never on the whole, and never add denominators when adding unlike fractions. Do not write on this paper — mark answers on your answer sheet.

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1. Compute 1 1/4 × 2 2/5 × 1 1/6 and give the answer as a mixed number.

(A) 3 1/2 (B) 3 11/120
(C) 2 1/2 (D) 4 3/10

2. A barrel is 4/5 full. First 1/4 of the water in it is drawn off, then 1/3 of what then remains is drawn off. What fraction of the FULL barrel's capacity is now in it?

(A) 7/15 (B) 1/5
(C) 2/5 (D) 8/15

3. Compute 3 3/4 ÷ 1 1/2 ÷ 2 1/2.

(A) 1 (B) 25/4
(C) 9/16 (D) 5/2

4. A man spends 2/5 of his salary on rent, then 1/3 of the remainder on food, and is then left with Rs 4800. What was his salary in rupees?

(A) 18000 (B) 12000
(C) 8000 (D) 11520

5. Which of these four fractions is the LARGEST: 7/9, 5/6, 11/15, 13/18?

(A) 7/9 (B) 11/15
(C) 13/18 (D) 5/6

6. Compute (5/6 – 1/4) × (2/3 + 1/2).

(A) 7/13 (B) 49/72
(C) 35/72 (D) 1/2

7. Compute (2 1/4 × 1 1/3) ÷ 1 1/5.

(A) 2 1/2 (B) 3 3/5
(C) 9/16 (D) 2 1/4

8. On a number line, what fraction lies exactly one-third of the way from 1/4 to 5/8?

(A) 3/8 (B) 1/2
(C) 5/16 (D) 11/24

9. Of a herd, 3/8 are cows. Of the cows, 2/3 are brown; of the non-cows, 1/4 are brown. What fraction of the WHOLE herd is brown?

(A) 11/24 (B) 13/32
(C) 1/4 (D) 5/12

10. Without computing exactly, which statement about 30 ÷ 6/7 is TRUE?

(A) It equals 30. (B) It is less than 30 but more than 25.
(C) It is less than 6/7. (D) It is greater than 30.

11. After giving away 3/7 of his stickers, then 1/4 of the remainder, then 1/3 of what was then left, Karan has 16 stickers. How many did he start with?

(A) 42 (B) 64
(C) 56 (D) 48

12. Arrange in ASCENDING order: 7/12, 5/8, 11/18. Which order is correct?

(A) 11/18 < 7/12 < 5/8 (B) 7/12 < 11/18 < 5/8
(C) 5/8 < 11/18 < 7/12 (D) 7/12 < 5/8 < 11/18

13. A vessel is 3/5 full. After 8 litres are poured in, it is 13/15 full. What is the vessel's full capacity in litres?

(A) 24 (B) 27
(C) 36 (D) 30

14. Compute 5/6 of (2 2/5 ÷ 1 1/3) and give the answer as a mixed number.

(A) 1 1/2 (B) 1 4/5
(C) 2 3/5 (D) 9/10

15. Compute (3/4 + 5/6) – (2/3 – 1/4).

(A) 2 1/4 (B) 5/6
(C) 1 1/6 (D) 13/12

16. A tank is filled by tap A to 5/12 of its volume and by tap B to 3/8 of its volume in one hour together. In the next hour 1/3 of the water then present is drained. What fraction of the full tank remains?

(A) 13/24 (B) 19/36
(C) 19/24 (D) 7/12

17. By what number must 2 1/4 be divided to give 3/5?

(A) 27/20 (B) 4/15
(C) 5/12 (D) 3 3/4

18. A rope is 5 1/4 m long. Pieces of 7/8 m are cut from it until no full piece remains. How many full pieces are cut and how much rope is left over?

(A) 5 pieces, 7/8 m left (B) 6 pieces, 7/8 m left
(C) 7 pieces, 0 m left (D) 6 pieces, 0 m left

19. Half of one-fifth of a quantity equals one-third of another quantity. If the first quantity is 90, what is the second quantity?

(A) 54 (B) 9
(C) 27 (D) 45

20. Compute 1 1/3 ÷ 2/5 ÷ 1 2/3 and give the answer as a mixed number.

(A) 2 (B) 8/9
(C) 5/9 (D) 100/9

21. Two pipes fill a tank: pipe A fills 2/7 of it per hour, pipe B fills 5/14 of it per hour. Working together, how many hours to fill the whole tank?

(A) 9/14 (B) 14/9
(C) 9/7 (D) 7/2

22. Compute 4 1/2 × 2 2/3 ÷ 3 3/5.

(A) 43 1/5 (B) 3 1/3
(C) 27/40 (D) 12

23. A field is 6/7 of an acre. One-half of it is for maize, and 3/5 of the REMAINING field is for cotton. What fraction of an acre is for cotton?

(A) 18/35 (B) 3/7
(C) 9/14 (D) 9/35

24. By what number must 5/8 be multiplied so the product becomes 1 7/8?

(A) 75/64 (B) 1 1/4
(C) 3 (D) 8/15

25. A jar of nuts is shared so the first child takes 1/4, the second takes 1/3 of what is left, and the third takes 2/5 of what then remains. If the third child gets 8 nuts, how many nuts were in the jar?

(A) 20 (B) 40
(C) 30 (D) 60

26. Compute (2/3 ÷ 4/9) × (3/10 ÷ 9/5).

(A) 4 (B) 1/4
(C) 27/100 (D) 1/16

27. Compute 7/8 of 16/21 of 3/4 of 56.

(A) 28 (B) 21
(C) 32 (D) 14

28. Compute 5 1/3 ÷ 1 1/9 and write it as a mixed number.

(A) 4 4/5 (B) 5 23/27
(C) 5/6 (D) 4 2/3

29. Two-ninths of a pole is buried in mud and 1/3 of the pole is under water; 20 m sticks out above the water. How long is the pole in metres?

(A) 45 (B) 36
(C) 30 (D) 60

30. Compute 9/10 ÷ (3/4 × 2/5).

(A) 27/100 (B) 10/3
(C) 3 (D) 3/10

31. A book has 480 pages. Sana reads 5/12 on the first day, then 2/7 of the remaining on the second day. How many pages are unread after two days?

(A) 280 (B) 240
(C) 180 (D) 200

32. Half of two-fifths of a number equals 24. What is three-quarters of the same number?

(A) 45 (B) 120
(C) 18 (D) 90

33. Compute 1/3 of 6 ÷ 1/2 of 8.

(A) 32 (B) 1/2
(C) 8 (D) 2

34. A 3 1/3 kg bag of sugar is repacked into 5/12 kg pouches. How many full pouches are made and what weight (in kg) is left over?

(A) 7 pouches, 5/12 kg left (B) 8 pouches, 5/12 kg left
(C) 8 pouches, 0 kg left (D) 6 pouches, 1/3 kg left

35. Which is larger: 4/5 of 7/8, or 7/8 ÷ 4/5?

(A) 4/5 of 7/8 is larger, equal to 7/10 (B) they are equal at 7/10
(C) 7/8 ÷ 4/5 is larger, equal to 35/32 (D) 4/5 of 7/8 is larger, equal to 35/32

36. A bottle holds 2 1/4 litres. First 1/3 of its contents is poured out, then 2/5 of what remains is poured out. How many litres remain in the bottle?

(A) 3/4　　(B) 27/20
(C) 6/5　　(D) 9/10

37. Compute (2 1/2 + 1 1/3) ÷ (1 1/6 – 2/3).

(A) 7 2/3　　(B) 2 5/18
(C) 23/6　　(D) 1/7

38. A worker does 2/9 of a job on day 1. On day 2 he does 1/3 of the remaining job; on day 3 he does 1/2 of what is then left. What fraction of the whole job is still undone after day 3?

(A) 20/27　　(B) 7/27
(C) 1/2　　(D) 5/27

39. A photo 5/6 m wide is enlarged to 2 1/4 times its width. By how much, in metres, does the new width exceed 1 1/2 m?

(A) 3/8　　(B) 15/8
(C) 5/12　　(D) 11/24

40. Compute 1 7/8 ÷ 3/4 ÷ 1 1/4 and write it as a mixed number.

(A) 25/32　　(B) 1 9/16
(C) 2　　(D) 2 13/32

41. In a crate, 4/9 of the fruits are mangoes, 1/3 are guavas, and the remaining 60 are bananas. How many fruits are in the crate?

(A) 540　　(B) 270
(C) 180　　(D) 135

42. A tank leaks 1/5 of its current water every hour. If it starts full, what fraction of the original water remains after 3 hours?

(A) 2/5　　(B) 61/125
(C) 64/125　　(D) 1/125

43. Compute 3 1/5 × 1 7/8 × 5/12 and write it as a mixed number.

(A) 3 1/3　　(B) 6
(C) 1 7/8　　(D) 2 1/2

44. A recipe uses 3/4 cup of oats for every 2/5 cup of honey. How many cups of oats are needed for 1 1/2 cups of honey?

(A) 4/5　　(B) 9/20
(C) 3 3/4　　(D) 2 13/16

45. Tap A fills 5/9 of a tank in 1 hour; tap B empties 1/6 of the tank in 1 hour. With both open, what fraction of the tank fills in 1 hour?

(A) 13/18 (B) 7/18
(C) 23/18 (D) 4/15

46. A 4 1/2 litre paint can covers 3/5 of a room with one coat. How many litres are needed to give the WHOLE room one coat?

(A) 2 7/10 (B) 75/32
(C) 7 1/2 (D) 6

47. Three-eighths of a number is 6 more than one-fourth of the same number. What is the number?

(A) 48 (B) 96
(C) 24 (D) 16

48. Compute (7/8 – 1/3) ÷ (5/6 – 1/4).

(A) 13/14 (B) 14/13
(C) 1 (D) 5/8

49. A merchant has 15 3/4 m of cloth. He sells 4/9 of it, then 2/5 of the remainder. How many metres are left?

(A) 5 1/4 (B) 8 3/4
(C) 7 (D) 3 1/2

50. Compute 6 ÷ 2/5 ÷ 3 ÷ 5/4.

(A) 9/100 (B) 25/4
(C) 4 (D) 1/4

51. A school has 720 students. 7/12 are in the senior section. Of the seniors, 3/7 take music; of those music students, 1/2 also take art. How many seniors take both music and art?

(A) 180 (B) 420
(C) 60 (D) 90

52. Which expression has the LARGEST value: 12 × 3/8, 12 ÷ 8/3, 12 × 7/4, or 12 ÷ 3/8?

(A) 12 × 7/4 (B) 12 ÷ 8/3
(C) 12 × 3/8 (D) 12 ÷ 3/8

53. On a number line, A is at 2/3 and B is at 11/6. A point P divides AB so that AP is 2/5 of AB. Where is P?

(A) 5/4 (B) 17/15
(C) 7/15 (D) 37/30

54. Compute 2 1/4 ÷ (1 1/2 ÷ 2/3).

(A) 81/16 (B) 1
(C) 9/16 (D) 2 1/4

55. Which statement is FALSE?

(A) Dividing a number by a proper fraction gives a result larger than the number.

(B) Multiplying a number by a proper fraction gives a smaller result.

(C) Dividing a number by a fraction greater than 1 always gives a result larger than the number.

(D) The reciprocal of a fraction greater than 1 is less than 1.

56. A vessel is 3/8 full. After 14 litres are added it becomes 5/8 full. What is the full capacity in litres?

(A) 56
(B) 28
(C) 70
(D) 112

57. Compute 2 2/3 ÷ 1 1/5 ÷ 5/9 and write it as a mixed number.

(A) 20/27
(B) 32/27
(C) 1 1/9
(D) 4

58. A field is split into four parts. The first is 1/4 of the field, the second is 1/6, the third is 1/3, and the fourth is the rest. The fourth part is what fraction of the field?

(A) 3/4
(B) 1/12
(C) 1/4
(D) 5/12

59. A juice mix is 2/9 mango pulp, 1/6 orange pulp, and the rest water. If the mix uses 110 mL of water, how many mL is the whole mix?

(A) 360
(B) 90
(C) 180
(D) 198

60. Compute 2/3 of 1 4/5 of 5/6 of 2 1/4 and give the answer as a mixed number.

(A) 1 7/8
(B) 3 3/5
(C) 27/16
(D) 2 1/4

Solutions — Mathematics (Class 7), Chapter 8: Working with Fractions — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	B	31	D	41	B	51	D
2	C	12	B	22	B	32	D	42	C	52	D
3	A	13	D	23	D	33	B	43	D	53	B
4	B	14	A	24	C	34	C	44	D	54	B
5	D	15	C	25	B	35	C	45	B	55	C
6	B	16	B	26	B	36	D	46	C	56	A
7	A	17	D	27	A	37	A	47	A	57	D
8	A	18	D	28	A	38	B	48	A	58	C
9	B	19	C	29	A	39	A	49	A	59	C
10	D	20	A	30	C	40	C	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Convert all: 5/4 × 12/5 × 7/6. Multiply: (5×12×7)/(4×5×6) = 420/120 = 7/2 = 3 1/2. A common slip is keeping the whole-number parts separate (1×2×1 = 2 plus fractions) which loses the cross terms; the only correct route is converting to improper fractions first.

2. (C) Start 4/5. After removing 1/4: remaining = 4/5 × 3/4 = 3/5. After removing 1/3 of that: 3/5 × 2/3 = 2/5. Each 'of what remains' must multiply the leftover, not the original 4/5; subtracting the fractions of the whole instead gives the wrong leftover.

3. (A) 15/4 ÷ 3/2 ÷ 5/2. Left to right: 15/4 × 2/3 = 30/12 = 5/2; then 5/2 × 2/5 = 10/10 = 1. Only the divisor is flipped at each step; flipping the dividend or chaining the division in the wrong order produces a different value.

4. (B) After rent, 3/5 left. After food, 2/3 of that remains: 3/5 × 2/3 = 2/5 of salary = Rs 4800. So salary = 4800 × 5/2 = Rs 12000. Treating the 1/3 as 1/3 of the whole salary (not of the remainder) misvalues what is left.

5. (D) Use denominator 90: $7/9 = 70/90$, $5/6 = 75/90$, $11/15 = 66/90$, $13/18 = 65/90$. The largest numerator over 90 is $75/90 = 5/6$. Comparing only numerators or only denominators without a common base misranks them.

6. (B) $5/6 - 1/4 = 10/12 - 3/12 = 7/12$. $2/3 + 1/2 = 4/6 + 3/6 = 7/6$. Product: $7/12 \times 7/6 = 49/72$, already in lowest terms. Adding numerators and denominators straight across when summing the bracket (e.g. $7/13$) is the classic error to avoid.

7. (A) $9/4 \times 4/3 = 36/12 = 3$; then $3 \div 6/5 = 3 \times 5/6 = 15/6 = 5/2 = 2\ 1/2$. Multiplying by $1\ 1/5$ instead of dividing (i.e. not flipping the divisor) gives $3 \times 6/5 = 18/5 = 3\ 3/5$, the trap value.

8. (A) Distance $= 5/8 - 1/4 = 5/8 - 2/8 = 3/8$. One-third of it $= 3/8 \times 1/3 = 1/8$. Point $= 1/4 + 1/8 = 2/8 + 1/8 = 3/8$. Taking the midpoint ($1/2$ the gap) instead of one-third gives $7/16$, and one-third of the way being confused with one-third of $5/8$ gives wrong values.

9. (B) Cows brown: $3/8 \times 2/3 = 6/24 = 1/4$ of herd $= 8/32$. Non-cows $= 5/8$; their brown share: $5/8 \times 1/4 = 5/32$ of herd. Total brown $= 8/32 + 5/32 = 13/32$. Simply averaging $2/3$ and $1/4$ ignores the unequal group sizes.

10. (D) Dividing by a fraction less than 1 enlarges the number, so $30 \div 6/7 = 30 \times 7/6 = 35 > 30$. Confusing division by a proper fraction with multiplication by it (which would shrink to under 30) gives the wrong direction.

11. (C) Fraction left $= 4/7 \times 3/4 \times 2/3 = (4\times3\times2)/(7\times4\times3) = 24/84 = 2/7$. So $2/7$ of start $= 16$, start $= 16 \times 7/2 = 56$. Each 'of the remainder' must keep multiplying the running leftover; subtracting the named fractions from 1 separately mis-states the leftover.

12. (B) LCM of 12, 8, 18 is 72: $7/12 = 42/72$, $5/8 = 45/72$, $11/18 = 44/72$. Ascending: $42 < 44 < 45$, i.e. $7/12 < 11/18 < 5/8$. Ranking by numerator size ($7 > 5$) or denominator size alone gives a wrong order.

13. (D) Rise $= 13/15 - 3/5 = 13/15 - 9/15 = 4/15$ of capacity $= 8$ litres. Capacity $= 8 \times 15/4 = 30$ litres. Subtracting the fractions wrongly (e.g. $13/15 - 3/5$ read as $13/15 - 3/15$) gives $10/15$ and a wrong capacity.

14. (A) $2\ 2/5 = 12/5$, $1\ 1/3 = 4/3$. Divide: $12/5 \times 3/4 = 36/20 = 9/5$. Then $5/6$ of $9/5 = 5/6 \times 9/5 = 45/30 = 3/2 = 1\ 1/2$. Multiplying instead of dividing inside the bracket ($12/5 \times 4/3$) breaks the result.

15. (C) $3/4 + 5/6 = 9/12 + 10/12 = 19/12$. $2/3 - 1/4 = 8/12 - 3/12 = 5/12$. Difference: $19/12 - 5/12 = 14/12 = 7/6 = 1\ 1/6$. Dropping the brackets and combining left-to-right with sign errors gives a different total.

16. (B) Filled = 5/12 + 3/8 = 10/24 + 9/24 = 19/24. Drained 1/3, so 2/3 remains: 19/24 × 2/3 = 38/72 = 19/36. Draining 1/3 of the full tank (not of the water present) over-removes water.

17. (D) We need 9/4 ÷ x = 3/5, so x = (9/4) ÷ (3/5) = 9/4 × 5/3 = 45/12 = 15/4 = 3 3/4. Multiplying 9/4 by 3/5 instead of dividing gives 27/20, the trap; the unknown divisor is found by dividing dividend by quotient.

18. (D) 5 1/4 = 21/4 m, each piece 7/8 m. 21/4 ÷ 7/8 = 21/4 × 8/7 = 168/28 = 6 exactly. So 6 full pieces and nothing left. Reading 21/4 ÷ 7/8 as 21/4 × 7/8 (no flip) gives a fraction under 6 and a false leftover.

19. (C) Half of 1/5 of 90 = 1/2 × 1/5 × 90 = 90/10 = 9. This equals 1/3 of the second, so second = 9 × 3 = 27. Forgetting the 'half' (using only 1/5 of 90 = 18, then ×3 = 54) is the trap.

20. (A) 4/3 ÷ 2/5 = 4/3 × 5/2 = 20/6 = 10/3; then 10/3 ÷ 5/3 = 10/3 × 3/5 = 30/15 = 2. Each division flips only its own divisor; collapsing both divisors at once or skipping a flip changes the answer.

21. (B) Combined rate = 2/7 + 5/14 = 4/14 + 5/14 = 9/14 of the tank per hour. Time = 1 ÷ 9/14 = 14/9 hours. Adding the times instead of the rates, or forgetting to take the reciprocal of the combined rate, gives 9/14.

22. (B) 9/2 × 8/3 = 72/6 = 12; then 12 ÷ 18/5 = 12 × 5/18 = 60/18 = 10/3 = 3 1/3. Multiplying by 3 3/5 instead of dividing (12 × 18/5) gives 43 1/5, the trap.

23. (D) Field = 6/7. Maize = 1/2 × 6/7 = 3/7; remaining = 3/7. Cotton = 3/5 × 3/7 = 9/35 acre. Applying 3/5 to the whole 6/7 (not to the 3/7 remaining) gives 18/35, the trap.

24. (C) Need 5/8 × x = 15/8, so x = 15/8 ÷ 5/8 = 15/8 × 8/5 = 120/40 = 3. Multiplying 5/8 by 15/8 instead of dividing gives 75/64, the trap; the missing factor is target divided by the known factor.

25. (B) After first: 3/4 left. After second (1/3 of that): 3/4 × 2/3 = 1/2 left. Third takes 2/5 of 1/2 = 1/5 of jar = 8 nuts, so jar = 8 × 5 = 40. Treating each fraction as of the whole jar mis-sizes the third child's share.

26. (B) 2/3 ÷ 4/9 = 2/3 × 9/4 = 18/12 = 3/2. 3/10 ÷ 9/5 = 3/10 × 5/9 = 15/90 = 1/6. Product: 3/2 × 1/6 = 3/12 = 1/4. Flipping the wrong term in either division reverses the size of each factor.

27. (A) 'of' means multiply: 7/8 × 16/21 × 3/4 × 56. Step by step: 7/8 × 56 = 49; 49 × 16/21 = 784/21 = 112/3; 112/3 × 3/4 = 336/12 = 28. Reordering correctly: the exact value is 28. Mistaking any 'of' for division would shrink or grow the chain wrongly.

28. (A) $5\ 1/3 = 16/3$, $1\ 1/9 = 10/9$. Divide: $16/3 \times 9/10 = 144/30 = 24/5 = 4\ 4/5$. Forgetting to convert the mixed numbers (e.g. $5 \div 1$ dealt with separately) destroys the result.

29. (A) Hidden fraction = $2/9 + 1/3 = 2/9 + 3/9 = 5/9$. Above = $1 - 5/9 = 4/9$ of pole = 20 m. Pole = $20 \times 9/4 = 180/4 = 45$ m. Adding the buried fractions with mismatched denominators (e.g. 3/12) mis-states the above-water fraction.

30. (C) Bracket first: $3/4 \times 2/5 = 6/20 = 3/10$. Then $9/10 \div 3/10 = 9/10 \times 10/3 = 90/30 = 3$. Dividing before doing the bracket, or flipping 9/10 instead of the divisor, gives a wrong value.

31. (D) Day 1: $5/12 \times 480 = 200$ read, 280 left. Day 2: $2/7 \times 280 = 80$ read, $280 - 80 = 200$ unread. Taking 2/7 of the whole 480 (not of the 280 remaining) over-counts the second day's reading.

32. (D) $1/2 \times 2/5 \times N = 24$, so $1/5 \times N = 24$, $N = 120$. Then 3/4 of 120 = 90. Forgetting the 'half' (using $2/5 \times N = 24$, $N = 60$) gives a wrong N and a trap of 45.

33. (B) Read 'of' as multiply, evaluate the products first: (1/3 of 6) ÷ (1/2 of 8) = $2 \div 4 = 1/2$. Treating 'of' as a plain ÷ chain left-to-right, e.g. $1/3 \times 6 \div 1/2 \times 8 = 32$, is the classic precedence trap.

34. (C) $3\ 1/3 = 10/3$ kg; $10/3 \div 5/12 = 10/3 \times 12/5 = 120/15 = 8$ exactly. So 8 full pouches, nothing left. Dividing without flipping ($10/3 \times 5/12$) gives a small fraction and a false leftover.

35. (C) 4/5 of 7/8 = $4/5 \times 7/8 = 28/40 = 7/10$. $7/8 \div 4/5 = 7/8 \times 5/4 = 35/32$, which is more than 1 and so larger than 7/10. Dividing by a proper fraction (4/5) enlarges, while multiplying by it shrinks; assuming both give the same value is the trap.

36. (D) Full 9/4 L. After 1/3 out, 2/3 remain: $9/4 \times 2/3 = 18/12 = 3/2$. After 2/5 out, 3/5 remain: $3/2 \times 3/5 = 9/10$ L. Subtracting 1/3 and 2/5 as fractions of the whole bottle over-removes liquid.

37. (A) Sum: $5/2 + 4/3 = 15/6 + 8/6 = 23/6$. Difference: $7/6 - 2/3 = 7/6 - 4/6 = 3/6 = 1/2$. Divide: $23/6 \div 1/2 = 23/6 \times 2/1 = 46/6 = 23/3 = 7\ 2/3$. Not flipping the 1/2 divisor (multiplying by 1/2) halves the result instead.

38. (B) After day 1: 7/9 left. Day 2 does 1/3 of 7/9, leaving $2/3 \times 7/9 = 14/27$. Day 3 does 1/2 of that, leaving $1/2 \times 14/27 = 7/27$ undone. Applying each fraction to the whole job rather than the running remainder mis-tracks the leftover.

39. (A) New width = $5/6 \times 9/4 = 45/24 = 15/8$ m. Excess over $3/2 = 15/8 - 12/8 = 3/8$ m. Forgetting to subtract the $1\ 1/2$ (just giving 15/8) or comparing to the original width is the trap.

40. (C) $15/8 \div 3/4 = 15/8 \times 4/3 = 60/24 = 5/2$; then $5/2 \div 5/4 = 5/2 \times 4/5 = 20/10 = 2$. Flipping the dividend instead of each divisor, or chaining out of order, gives a wrong product.

41. (B) Mangoes + guavas = $4/9 + 1/3 = 4/9 + 3/9 = 7/9$. Bananas = $2/9$ of crate = 60, so crate = $60 \times 9/2 = 270$. Adding denominators ($4/9 + 1/3$ read as $5/12$) mis-states the banana fraction.

42. (C) Each hour $4/5$ of the current water remains. After 3 hours: $(4/5)^3 = 64/125$. Subtracting $1/5$ three times from the whole ($1 - 3/5 = 2/5$) wrongly treats the leak as a fixed fraction of the original.

43. (D) $16/5 \times 15/8 \times 5/12 = (16\times15\times5)/(5\times8\times12) = 1200/480 = 5/2 = 2\ 1/2$. Keeping whole parts separate ($3\times1 = 3$ plus fraction bits) loses the cross products and misses the answer.

44. (D) Oats per cup of honey = $(3/4) \div (2/5) = 3/4 \times 5/2 = 15/8$ cups. For $3/2$ cups honey: $15/8 \times 3/2 = 45/16 = 2\ 13/16$ cups. Multiplying $3/4$ by $2/5$ (instead of dividing) to get the rate understates the oats needed.

45. (B) Net fill rate = $5/9 - 1/6 = 10/18 - 3/18 = 7/18$ per hour. Adding the rates (A plus B) instead of subtracting the drain over-counts the fill.

46. (C) Paint per whole room = $(9/2) \div (3/5) = 9/2 \times 5/3 = 45/6 = 15/2 = 7\ 1/2$ L. Multiplying $9/2$ by $3/5$ (instead of dividing) gives $27/10 = 2\ 7/10$, the trap.

47. (A) $3/8\ N - 1/4\ N = 6$. $3/8 - 2/8 = 1/8$, so $1/8\ N = 6$, $N = 48$. Subtracting the fractions wrongly ($3/8 - 1/4$ read as $2/4$) or dividing 6 by the wrong fraction gives the trap values.

48. (A) $7/8 - 1/3 = 21/24 - 8/24 = 13/24$. $5/6 - 1/4 = 10/12 - 3/12 = 7/12 = 14/24$. Divide: $13/24 \div 14/24 = 13/24 \times 24/14 = 13/14$. Flipping the dividend instead of the divisor inverts the answer to $14/13$.

49. (A) $63/4$ m. After selling $4/9$, remaining = $5/9 \times 63/4 = 315/36 = 35/4 = 8\ 3/4$. After selling $2/5$ of that, remaining = $3/5 \times 35/4 = 105/20 = 21/4 = 5\ 1/4$ m. Taking $2/5$ of the whole $63/4$ m, not of the remainder, over-sells.

50. (C) $6 \div 2/5 = 6 \times 5/2 = 15$; $15 \div 3 = 5$; $5 \div 5/4 = 5 \times 4/5 = 4$. Each division flips only its own divisor and is taken left to right; collapsing the chain or skipping a flip changes the value.

51. (D) Seniors = $7/12 \times 720 = 420$. Music = $3/7 \times 420 = 180$. Both = $1/2 \times 180 = 90$. Forgetting the final $1/2$ (stopping at 180) or applying the $3/7$ to all 720 mis-counts the group.

52. (D) $12 \times 3/8 = 9/2 = 4.5$; $12 \div 8/3 = 12 \times 3/8 = 9/2 = 4.5$; $12 \times 7/4 = 21$; $12 \div 3/8 = 12 \times 8/3 = 32$. The largest is 32 from $12 \div 3/8$: dividing by a small proper fraction enlarges the most. Treating $\div 3/8$ as $\times 3/8$ would instead shrink it to 4.5, the trap.

53. (B) $AB = 11/6 - 2/3 = 11/6 - 4/6 = 7/6$. $AP = 2/5 \times 7/6 = 14/30 = 7/15$. $P = 2/3 + 7/15 = 10/15 + 7/15 = 17/15$. Taking the midpoint (1/2 of AB) instead of 2/5, or forgetting to add A's position back, gives the trap values.

54. (B) Bracket: $3/2 \div 2/3 = 3/2 \times 3/2 = 9/4$. Then $9/4 \div 9/4 = 1$. Evaluating left-to-right without the bracket ($2\ 1/4 \div 1\ 1/2$ first) gives a different value; the bracketed division must be done first.

55. (C) Dividing by a fraction GREATER than 1 (e.g. $\div 5/4$) actually shrinks the number, so that claim is false. The other three are true: dividing by a proper fraction enlarges, multiplying by a proper fraction shrinks, and the reciprocal of an improper fraction is a proper fraction.

56. (A) Rise $= 5/8 - 3/8 = 2/8 = 1/4$ of capacity = 14 L. Capacity $= 14 \times 4 = 56$ L. Subtracting the fractions wrongly, or dividing 14 by the wrong fraction, mis-sizes the rise.

57. (D) $8/3 \div 6/5 = 8/3 \times 5/6 = 40/18 = 20/9$; then $20/9 \div 5/9 = 20/9 \times 9/5 = 180/45 = 4$. Flipping the wrong fraction at either step, or not converting the mixed numbers, breaks the chain.

58. (C) Used $= 1/4 + 1/6 + 1/3 = 3/12 + 2/12 + 4/12 = 9/12 = 3/4$. Rest $= 1 - 3/4 = 1/4$. Adding denominators or using a wrong common denominator gives a false used-fraction and a wrong remainder.

59. (C) Pulp fraction $= 2/9 + 1/6 = 4/18 + 3/18 = 7/18$. Water $= 11/18$ of mix = 110 mL, so mix $= 110 \times 18/11 = 180$ mL. Adding the pulp fractions with mismatched denominators (e.g. 3/15) mis-sizes the water share.

60. (D) All 'of' mean multiply: $2/3 \times 9/5 \times 5/6 \times 9/4 = (2\cdot9\cdot5\cdot9)/(3\cdot5\cdot6\cdot4) = 810/360 = 9/4 = 2\ 1/4$. Mistaking any 'of' for division, or keeping the whole-number parts separate, would change the chain.